Milestone 2

https://github.com/com-480-data-visualization/special-operations/

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April 2026

1 Project Goal

1.1 Motivation

For a very long time, Europe was the economic powerhouse of the world. However, recently, other regions seem to be picking up the pace and a narrative of decline around European development is getting stronger. It is essential to know the trajectory of the region, for citizens, policymakers, and entrepreneurs alike, to foresee where power will lie in the future if nothing changes. For our project, we intend to answer the question "Is Europe really falling behind?"

1.2 Visualization goal

The main aim of our visualizations is to convey the economic evolution of different regions and countries over time in order to make the answer to the abovementioned question clear.

2 Implementation Plan

2.1 MVP - Core Ideas

The core visualization is an interactive map where users can choose an economic indicator and the countries will be colored accordingly. To show the evolution through time, the map will be normalized at a referenced year, and animated to step through the years. We will explore alternate complementary visualization options like a scatter plot to make differences clearer.

Tool: D3. Lecture: 8 Maps.

While the core visualization focuses on a broad overview of the world with respect to a single chosen metric, we plan to add additional visualizations to make more direct comparisons between countries and regions across metrics. To achieve this, we will use a spider graph, with the axes being different social and economic indicators. The user will be able to choose a (small) set of countries

to overlap and visualize, alongside a reference year at which the spider graph is normalized, making the differences in evolution at different points apparent and easily digestible.

Tool: D3. Lecture: 11 Tabular Data.

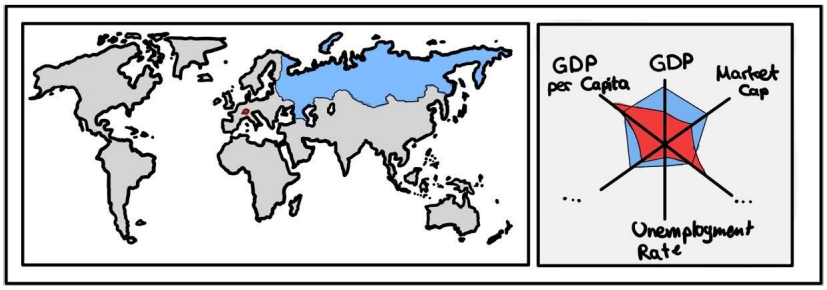

Figure 1: Spider Graph Comparison: A multi-axis visualization comparing key economic indicators (e.g., Tech Share, GDP Growth, Unemployment Rate) between regions like the USA and Europe to highlight structural divergence.

2.2 Possible enhancements

To make the visualization more intuitive, we plan to make a cartogram, where the size of the countries will shift according to the evolution of the chosen metric. It makes big differences stand out more and would make the visualization more appealing.

Tool: Topogram library for d3: https://github.com/shawnbot/topogram.

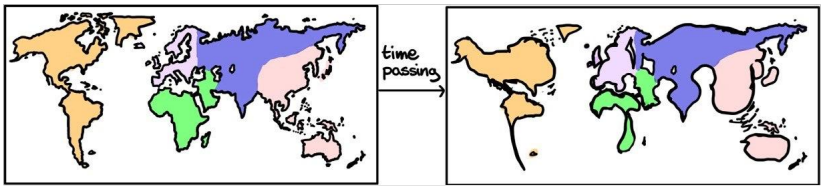

Figure 2: Dynamic Cartogram: This visualization distorts country sizes based on economic metrics (e.g., GDP or Market Cap), illustrating the shifting global economic "gravity" over time.

Continuing the trend of magnification, we plan to explore the evolution of a single country as split into different contributing factors and/or fields. This would make the analysis more comprehensive by helping understand what industries are driving the (relative) economic performance in a region. To achieve this, we plan to use a tree map.

Tool: D3. Lecture: 10 Graphs.

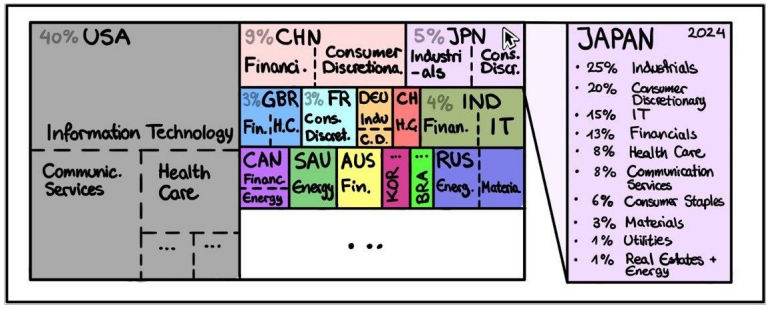

Figure 3: Tree map representing the industry breakdown of a specific country (e.g., Japan 2024), illustrating the dominant sectors like Industrials and Consumer Discretionary.